

CARGOMULE · DESIGN DECISIONS REGISTER

CargoMule design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cargomule		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the decision records, the review note and the build plan work

CargoMule design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Response to the R8 miss: the constructable trailer weighs 46.2 kg against 45 kg	(a) relax R8 to 47 kg for the first prototype and weigh it at TRL 4; (b) straps or mesh in place of the plywood side boards now (about 43.9 kg); (c) find about 0.5 kg elsewhere	(a)	Side boards (build plan section 3.9) under (b); none under (a)	CGM-DDR-003, A1
2	Where the key switch and charge port go, now that the enclosure's front face is its door	(a) enclosure right wall near the front; (b) on the door with a lead across the hinge	(a)	Enclosure drilling (section 3.10), harness	CGM-DDR-003, A2
3	Coupler damping	(a) friction washers on the lever pivot; (b) a small hydraulic damper between the cheeks and the lever	(a) for the prototype; check for chatter at TRL 4	Brake lever (section 3.6)	CGM-DDR-003, A3
4	Bringing the appearance model and photoreal renders up to the constructable design	(a) update on Amish's Mac; (b) leave until TRL 4	(a), before the repo is made public	None; media only	CGM-DDR-003, A4
5	Finish and small parts on the renders: graphite powder coat, faced boards with badges, deck tie-down tracks, coupler gaiter, drawbar reflector	Keep the BOM's paint and plywood for the prototype, or adopt the styling	Keep the BOM for the prototype; treat the rest as finished-product styling	None for the prototype	Review note 2026-09-26, item 4
6	Hero render composition: the trailer fills about half the frame	Accept, or add a close-up camera option to the kit renderer	Accept	None	Review note 2026-09-26, item 5
7	First users and region for co-design	Market traders, a trades cooperative or a community food bank	None; partners to be picked per area later	Not part of the TRL 3 build	CGM-DDR-001, O1
8	Road legality of a motorized bicycle trailer in the first target country	EU pedal-assist exclusion, US federal and state rules	None	Not part of the TRL 3 build; needed before any public road use	CGM-DDR-001, O2

The review note of 2026-09-26 also proposed a closed enclosure with a bottom or front access panel (item 1) and a clear guard over the load cell (item 3). Both are overtaken by CGM-DDR-003: the enclosure now has a front door and the load cell sits inside the coupler housing.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The S-type load cell fits the 56.3 mm housing bore: body no more than 54 mm across its diagonal (about 51 x 13 x 61 mm assumed), M10 threads at both ends	The cell lives inside the coupler housing	CGM-DDR-003, P1
2	The hub motor: 100 mm spacing, 10 mm axle flats, disc mount, shell no more than 156 mm across within 16 mm of the disc-side locknut, cable exit on the non-disc side, torque arm in the kit	Sets the slot width, the caliper clearance and the cable route	CGM-DDR-003, P9, P10
3	The caliper's post-mount adapter for a 180 mm rotor: hole spacing on the dropout tab	The tab's two holes are drilled to suit	CGM-DDR-003, P9
4	The hitch's arm is 32 mm or less and takes the 6 mm locking pin; the hitch fits the bicycle's axle type	The arm goes 60 mm into the drawbar's 33 mm bore	CGM-DDR-003, P7; R7
5	The preload spring: about 30 N installed, free length about 96 mm, OD 40 mm or less, solid length under 40 mm	It sets the brake's onset and must not go solid within the 50 mm stroke	CGM-CAL-001, F3
6	Motor constants (winding, resistance, thermal) from the datasheet	They decide whether R3 holds (decided under CGM-DDR-002; part selection is TRL 4)	CGM-DDR-002, D12
7	Rivet nuts rated for 1.5 mm wall box section	They carry the deck, the enclosure and the lights	CGM-DDR-003, P8, P11

■ Value engineering

Value-engineering target: USD 1,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 997 (USD 3 under the target).

Main cost drivers (CGM-CAL-001, I1): the battery pack (line 11, \$180), the hub motor wheel (line 7, \$160), the chassis frame (line 1, \$95), the deck and side boards (line 2, \$77) and the overrun brake coupler (line 6, \$58). The parts added to make the design buildable added \$18 (CGM-DDR-003).

Savings worth trying:

- Salvaged 20 in wheels and brakes and a lighter enclosure, which the TRL 3 review estimated could bring the parts to about \$900.
- Straps or mesh in place of the plywood side boards (about 43.9 kg), which also answers the R8 mass miss (CGM-DDR-003, A1).
- A SwapCell receiver in the fleet variant, which would price the shared pack once and leave it out of this kit (CGM-DDR-001, D2).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: cost cuts then a \$1,000 budget, 36 V LiFePO4 384 Wh pack, steel frame, one hub motor in the left wheel, proportional assist G = 2 to 6, overrun mechanical brake plus assist cut, pitch reworded, drawbar force sensing, 20 in wheels with the hitch on the left axle end, handcart mode out of scope	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	CGM-DDR-001
2026-09-25	TRL 3 items D11 to D14: R8 relaxed to 45 kg with straps or mesh as a later weight option, thermal derating and a use limit for long climbs, 180 mm rotors with metallic pads, gain range kept at 2 to 6	Amish: "i accept all your recommendations, go with them across all repos."	CGM-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan; outstanding decisions go in this register, not the plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CGM-DDR-003 (Draft, open for his review)